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ORIGINAL ARTICLE

Football (soccer) match-derived hamstring muscles residual fatigue can be monitored using early rate of torque development

Pedro L. Cosio[1] · Lia Moreno-Simonet[1] · Daniel Fernández[2] · Mario Lloret[1] · Xavier Padulles[1] · Josep M. Padulles[1] · Andreu Farran-Codina[3] · Gil Rodas[4,5] · Joan A. Cadefau[1,6]

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Abstract

Purpose The aim of this study was to determine whether a soccer match affects the rapid force-generating capacity of the hamstring muscles, given their key role in both horizontal ground reaction force production during sprint biomechanics, and in the deceleration of the shank during the late swing phase, where rapid force production is essential owing to time constraints. Therefore, the research objective was to determine soccer match-induced hamstrings residual fatigue and recovery through rate of torque development (RTD) and associated biochemical parameters.

Methods The recovery kinetics of hamstrings RTD metrics by the $90°_{hip}$:$20°_{knee}$ *test*, together with serum biomarkers (creatine kinase, mitochondrial creatine kinase, transaminases, malondialdehyde, irisin), were assessed in 19 male, regional first-division soccer players (age = 20.9 ± 2.0 years, mass = 72.6 ± 11.9 kg, height = 175.9 ± 6.9 cm [mean ± SD]), before a soccer match (MD) and post-24 h (MD_{+1}), post-48 h (MD_{+2}) and post-72 h (MD_{+3}), through a repeated measures design.

Results Early RTD to 50 ms ($p < 0.001$, $g = -1.24$) and 100 ms ($p < 0.001$, $g = -1.06$) remained unrecovered on MD_{+3} in both hamstring muscles. However, maximal voluntary isometric contraction (MVIC) torque of the dominant and non-dominant hamstrings was unrecovered on MD_{+2} ($p = 0.004$, $g = -0.91$; and $p = 0.002$, $g = -0.98$, respectively) and recovered on MD_{+3} ($p = 0.057$ and $p = 0.070$, respectively). Further, neuromuscular deficits were coupled with myocyte structural ($p = 0.002$, $g = 1.11$) and mitochondrial damage ($p = 0.004$, $g = 0.92$) biomarkers.

Conclusion Based in the findings, early $RTD_{0–50}$ and $RTD_{0–100}$ monitoring, through the $90°_{hip}$:$20°_{knee}$ *IPC test*, is a cost-effective method for assessing soccer match-induced hamstring muscles residual fatigue and recovery. Overall, soccer match-induced hamstring residual fatigue is not recovered within a 3-day recovery period. Practitioners can use rapid force production metrics through isometric assessments, providing a simple, non-exhaustive tool, for assessing residual fatigue status during congested competitive periods, to comprehensively balance muscle recovery with optimizing training.

Keywords Exercise-induced muscle damage · Force-generating capacity · Hamstring strain injury · Muscle recovery · Sprint-based sports

Abbreviations

Acc	Accelerations
ALT	Alanine aminotransferase
ANOVA	Analysis of variance

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✉ Joan A. Cadefau
jcadefau@gencat.cat

1 Institut Nacional d'Educació Física de Catalunya (INEFC), Universitat de Barcelona (UB), 08038 Barcelona, Spain

2 Sports Performance Department, Futbol Club Barcelona, Rink Hockey, 08970 Barcelona, Spain

3 Department of Nutrition, Food Science, and Gastronomy, INSA-UB, Faculty of Pharmacy and Food Sciences, Universitat of Barcelona (UB), 08921 Barcelona, Spain

4 Sports Medicine Unit, Hospital Clinic and Sant Joan de Déu, 08036 Barcelona, Spain

5 Medical Department, Medical Department of Futbol Club Barcelona (FIFA Medical Center of Excellence) and Barça Innovation Hub, 08970 Barcelona, Spain

6 Faculty of Medicine and Health Sciences, Universitat de Barcelona (UB), 08036 Barcelona, Spain

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AST	Aspartate aminotransferase
AUC	Area under the curve
BF_{LH}	Biceps femoris long head
Ca^{2+}	Calcium ion
CI	Confidence interval
CK	Creatine kinase
Dec	Decelerations
ES	Effect size
HSI	Hamstring strain injury
HSRd	High-speed running distance
ICC	Intraclass correlation coefficient
IMU	Inertial measurement unit
IPC	Isometric posterior chain
MD	Match day
MDA	Malondialdehyde
MVIC	Maximal voluntary isometric contraction
RTD	Rate of torque development
SD	Standard deviation
sMtCK	Sarcomeric mitochondrial CK
SWC	Smallest worthwhile change
TD	Total distance
TE	Typical error

Introduction

Soccer is a team sport with intermittent, high-intensity repeated activities, including sprinting, accelerations or changes of direction (Gualtieri et al. 2023). Although differences may exist across playing positions (Miguel et al. 2022) or tactical formations (Modric et al. 2020), match running performance is similar between soccer players at different competitive standards (Lago-Peñas et al. 2024), exceeding a total distance (TD) covered of 10,000 m (Miguel et al. 2022; Lago-Peñas et al. 2023). However, covering this distance in 90 min should not be fatiguing for an athlete; rather, repeated high-intensity efforts, including high-speed running (i.e., >21 km h^{-1}) distance (HSRd), are the main source of fatigue, exceeding 400 m during a soccer match (Asian-Clemente et al. 2021; Lago-Peñas et al. 2023; Gualtieri et al. 2023; Rabano-Munoz et al. 2024). Furthermore, high-intensity efforts (e.g., HSRd, number of high-intensity sprints) have been significantly increasing during the last seasons (from 2012 to 2020) of the *Spanish LaLiga* in all playing positions by 6.1% and 11.3%, respectively (Lago-Peñas et al. 2023). Consequently, hamstring strain injuries (HSI) showed a twofold increase over the last 20 years, predominantly located in the biceps femoris, due to repeated sprinting in soccer matches every 72–96 h (Ekstrand et al. 2023).

Hamstring muscles play a key role during high-speed running or sprinting, as it has been shown that there is a relationship between horizontal ground reaction force production and both, hamstrings eccentric torque and electromyographic activity, during the late swing phase of the running gait cycle (Morin et al. 2015). During the late swing phase, there is an extensive hip flexion with reduced knee flexion, leading to a maximum elongation of the hamstring muscle–tendon unit, while simultaneously undergoing an active lengthening eccentric hamstrings contraction (Thelen et al. 2005), constituting the main injury mechanism in HSIs (Danielsson et al. 2020). Specifically, HSIs occur in a temporal period of less than 130 ms during the late swing phase (Heiderscheit et al. 2005). The biceps femoris long head (BF_{LH}) has been found to be the most important hamstring muscle during the late swing phase. The reason is that the lower limbs positioning during the late swing phase causes a greater BF_{LH} stretching and moment arm (Thelen et al. 2005), electromyographic activity (Higashihara et al. 2010) and musculotendinous strains (Schache et al. 2012), compared to the semitendinosus or semimembranosus muscles, and explains why BF_{LH} strains are the most common HSI in sprint-based team sports in both males and female athletes (Gudelis et al. 2024). In fact, it has recently been shown that moderate BF_{LH} fiber disruptions induced by a soccer match can be persistent up to 72 h (Carmona et al. 2024a). Hence, eccentric lengthening contractions during repeated sprints induce non-overlapping sarcomeres and failures in the excitation–contraction coupling process and, subsequently, long-lasting reductions in hamstring's force-generating capacity owing to ultrastructural damage (i.e., myofibrillar disruptions) and membrane permeability (Raastad et al. 2010; Hyldahl and Hubal 2014).

Returning to high-intensity intermittent exercise before complete muscle recovery aggravates neuromuscular impairments (Hubbard et al. 2023), which increases HSI risk (Green et al. 2020). Therefore, monitoring hamstring muscle recovery during congested competitive periods is therefore a key concern for practitioners to balance optimizing training with muscle recovery and decreasing HSI risk, requiring time- and cost-effective and non-exhaustive tests and outcomes. Although the most commonly used method for assessing hamstring's force-generating capacity is isokinetic testing, these devices present limitations that need to be considered (Asimakidis et al. 2024). Isokinetic testing is expensive, lacks portability and is time-consuming (Cuthbert et al. 2021; Asimakidis et al. 2024), so that clubs of lower competitive standards may not have access to this equipment due to financial requirements. However, isometric assessment through strain gauges or hand-held dynamometers are affordable solutions for clubs (Cuthbert et al. 2021), providing a fast, simple and cost-effective method with similar longitudinal construct validity (Knezevic et al. 2014) and high test–retest reliability (Moreno-Pérez et al. 2020) in both single- and multi-joint testing protocols. Accordingly, force-generating capacity loss is considered the most reliable indicator of muscle damage and recovery because it

is strongly correlated ($r = 0.89$) with the fraction of muscle fibers presenting ultrastructural alterations (Raastad et al. 2010). Hamstring's force-generating capacity has been commonly assessed using isometric posterior chain (IPC) supine tests. However, considering that lower limb positioning near to hip flexion (i.e., 90°) and knee extension (i.e., <30°) selectively activates the BF_{LH}, triggering the peak in the length–tension relationship (Chleboun et al. 2001) and the peak electromyographical activity (Kellis et al. 2017), Matinlauri et al. (2018) designed a new IPC standing test (i.e., *90°$_{hip}$:20°$_{knee}$ IPC test*), which has been found to be effective for monitoring hamstring's force-generating capacity after a sprint-based training (Cosio et al. 2024) or a soccer match (Carmona et al. 2024a).

The assessment of hamstring muscle recovery is required to consider the high angular velocity of the knee during the late swing phase, causing the hamstrings to apply rapid force to decelerate the shank before ground contact (Kalkhoven et al. 2023), which is the phase where the injury-related mechanism of HSI occurs (Danielsson et al. 2020). In particular, rapid force production is more sensitive than maximal force production to both training adaptations (Aagaard et al. 2002; Andersen et al. 2010) and acute responses to a single exercise session (Peñailillo et al. 2015; D'Emanuele et al. 2021). Therefore, it has been recommended to monitor rapid force production metrics (e.g., rate of torque development [RTD]) over early time intervals during post-exercise recovery period, to identify hamstring's neuromuscular deficits that increase HSI risk (D'Emanuele et al. 2021; Ripley et al. 2023, 2024a). Interestingly, it has been recently demonstrated that there is a significant decrease in hamstring's rapid force production immediately after a repeated sprint ability protocol (Bettariga et al. 2024a, b; Ripley et al. 2024b), while Grazioli et al. (2019) proved that RTD decrements immediately after a professional soccer match are greater than maximal voluntary isometric contraction (MVIC) torque decrements of the hamstring muscles. However, to the best of our knowledge, no previous studies have monitored the metrics of hamstring's rapid force production during the 72-h post-soccer match recovery period. Therefore, the purpose of the research was to monitor soccer match-induced hamstring's residual fatigue and recovery through RTD using a hamstring-specific neuromuscular test, together with associated serum biomarker recovery kinetics. We hypothesize that a soccer match triggers long-lasting decreases in hamstrings RTD metrics, accompanied by alterations in biomarkers of structural and oxidative damage.

Materials and methods

Participants

Nineteen male, regional first-division outfield soccer players (age = 20.9 ± 2.0 years, mass = 72.6 ± 11.9 kg, height = 175.9 ± 6.9 cm [mean ± SD]) accepted to voluntarily participate in the study. The playing history and injury records of the participants are shown in Table S1. Exclusion criteria were lower limb myotendinous injuries within 6 months prior to the experiment. The experiment was conducted during a 2-week break in the first regional division in which the players were competing. In this way, the players were able to rest for at least 3 days before the match exclusively scheduled for the experiment and during the 3 follow-up days after the match. All participants gave their written consent to participate in the study and confirmed that they received information on the experimental approach and associated risks. The experiment was conducted in accordance with the code of ethics of the World Medical Association (Declaration of Helsinki) and approved by the Ethics Committee for Clinical Research of the Catalan Sports Council (Generalitat de Catalunya) (037/CEICGC/2021). Sample size was computed a priori using G*Power software v3.1.9.6 based on MVIC change (*90°$_{hip}$:20°$_{knee}$ IPC test*) of the study by Matinlauri et al. (2018). A total of 19 participants were calculated to be included given an effect size (ES) of 0.68, a power of 80% and a 5% alpha error.

Experimental approach

The extent and time course of time-interval RTD to 50 ms ($RTD_{0–50}$), to 100 ms ($RTD_{0–100}$), to 250 ms ($RTD_{0–250}$) and peak RTD (RTD_{peak}), together with serum biomarkers (creatine kinase [CK], mitochondrial CK [sMtCK], transaminases, malondialdehyde [MDA] and irisin) were assessed before a soccer match (MD) and post-24 h (MD_{+1}), post-48 h (MD_{+2}) and post-72 h (MD_{+3}) through a single-group repeated measures design (Fig. 1). Participants completed the familiarization 1 week before the start of the experimental procedure, by repeating the neuromuscular tests until they were technically correct.

Participants were real-time monitored during an outfield soccer match (under current *FIFA laws of the game*) by means of validated 10-Hz inertial measurement unit (IMU) devices (*WIMU*™, RealTrack Systems, Almeria, Spain) (Bastida Castillo et al. 2018). The following thresholds were simultaneously monitored during the soccer match: TD > 10,000 m; HSRd > 400 m running over 21 km h^{-1}; accelerations (Acc), >30 accelerations over 3 m s^{-2}; and decelerations (Dec), >30 decelerations over 3 m s^{-2}.

Hamstring testing procedures

Maximal voluntary isometric contraction (MVIC) torque

Hamstrings MVIC torque was measured unilaterally by the *90°$_{hip}$:20°$_{knee}$ IPC test* (Matinlauri et al. 2018). This test provides strong validity in comparison to isometric gold

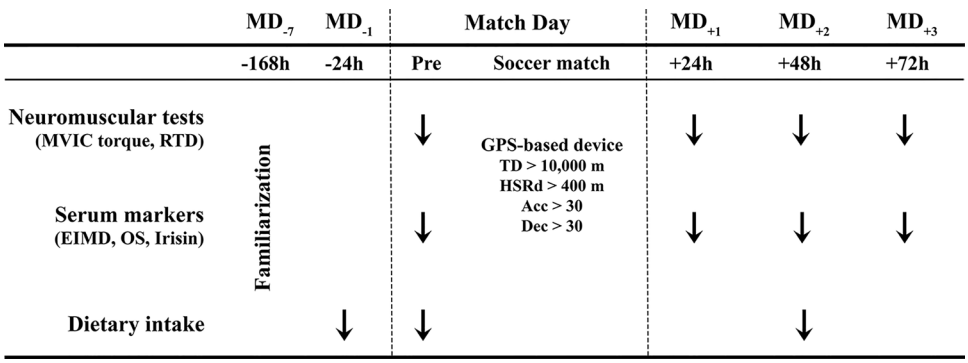

Fig. 1 Schematic overview showing the experimental approach. Acc, accelerations; Dec, decelerations; EIMD, exercise-induced muscle damage; GPS, global positioning system; HSRd, high-speed running distance; MD, match day; MVIC, maximal voluntary isometric contraction; OS, oxidative stress; RTD, rate of torque development; TD, total distance

standard dynamometry (Rasp et al. 2024) and "moderate to high" test–retest reliability (Matinlauri et al. 2018; Carmona et al. 2024b). Briefly, participants were verbally encouraged to apply force "as fast and hard as possible" to a rigid custom-built support attached to a strain gauge (Chronojump, Barcelona, Spain) and completed three trials of the test (3 s MVICs), with a 30-s rest between trials (Grazioli et al. 2019). MVIC was recorded as the 1-s window with the highest average force in the plateau of the signal, and the residual force resulting from the mass of the tested limb at rest was then subtracted (Cosio et al. 2024). Then, MVIC was multiplied by the individual length of the lever arm to calculate the moment of force (i.e., MVIC torque). The best of the three trials was recorded for further analyses. The used strain gauge is valid and reliable for peak and rapid force assessment (Courel-Ibáñez et al. 2020).

Rate of torque development (RTD)

Torque–time curves for each trial were analyzed using a standardized spreadsheet (Chavda et al. 2020) to compute $RTD_{0–50}$, $RTD_{0–100}$, $RTD_{0–250}$ and RTD_{peak}, that is, the steepest part of the curve. According to the recommendations of Maffiuletti et al. (2016): (1) the recording rigid custom-built strain gauge provides a low baseline noise amplitude (i.e., 0.148% MVIC torque); (2) the contraction onset was determined through a systematic-based method at the moment of "the last trough before force deflects above the range of the baseline noise"; (3) no filtering or signal smoothing was applied; and (4) average RTD of the three best trials was retained for further analysis. Also, each RTD metric was normalized to the respective MVIC torque ([RTD/MVIC torque] × 100) to monitor the ability to rapidly express the available force-generating capacity (Barber et al. 2024).

Blood sampling and processing

Blood samples were collected from an antecubital vein and then clotted at room temperature for 30 min in an SST II Advance tube (Becton Dickinson Vacutainer Systems, UK). Subsequently, samples were centrifuged at 4000×*g* for 8 min at 4 °C. After separation of the serum, aliquots were stored at −80 °C until analysis. Measurement of CK, alanine aminotransferase (ALT) and aspartate aminotransferase (AST) activities (intra-assay variability <5%; inter-assay variability <5%), which are widely used as muscle-specific biomarkers indicating sarcolemma disruption and permeability (Lim 2020), were conducted on an Advia 2400 automated device (Siemens™ Medical Solutions Diagnostics, Tarrytown, NY, USA), following the International Federation of Clinical Chemistry's reference procedures (Siekmann et al. 2002). sMtCK, as indicative of the structural integrity of muscle mitochondria (Ishikawa et al. 1997), was measured by ELISA kit #SEC386Hu (Cloud Clone Corp., Houston, TX, USA) according to the manufacturer's instructions (intra-assay variability <10%; inter-assay variability <12%). MDA, as the reference marker of free radical-induced lipid peroxidation (Spirlandeli et al. 2013), was measured by kit #E-EL-0060 (Elabscience, Wuhan, China) according to the manufacturer's instruction (intra-assay variability <10%; inter-assay variability <10%). The previously validated kit against mass spectrometry #EK-067-29 (Phoenix Pharmaceuticals Inc., Burlingame, CA, USA) was used to quantify circulating irisin (Ruan et al. 2019) according to manufacturer's instruction (intra-assay variability <10%; inter-assay variability <15%), owing to its protective effect against inflammatory and oxidative processes (Askari et al. 2018; Trettel et al. 2023) and promotion of muscle regeneration (Reza et al. 2017; Zhang et al. 2023).

Dietary intake record

The players were instructed on the procedures for recording their daily dietary intakes for 3 days (MD_{-1}, MD and MD_{+2}) and also provided with descriptors of commonly used household measurements (e.g., dessert spoon) and quantity measures (e.g., grams, millilitres), to verify that nutritional intake was homogeneous and within normal ranges for each macro and micronutrient. Then, the diet of each participant was calculated as the 3-day average through the dietary

analysis software PCN Pro v1. (Universitat de Barcelona, Barcelona, Spain). The volunteers were also provided with a standardized meal, composed of 200 g of pasta and 150 g of roasted chicken (total energy = 571 kcal, total carbohydrates = 44.4 g; total protein = 47.4 g; total lipids = 22.7 g), 2 h before the soccer match.

Statistical analyses

First, the intraclass correlation coefficient (ICC) was used to assess the reliability of a single data collector that examines the degree of concordance of scores between repetitions or sessions. The ICC was calculated for average measures using a two-way random effect model of absolute agreement. ICCs were interpreted based on the lower bound 95% CI as poor (<0.49), moderate (0.50–0.74), good (0.75–0.89) and excellent (>0.90) (Koo and Li 2016). Typical error (%TE) (Weir 2005) and smallest worthwhile change (SWC) (Hopkins 2004) were also calculated. Asymmetrically distributed variables were log- or square-transformed prior to analysis. A two-way repeated measures ANOVA (time [MD, MD_{+1}, MD_{+2}, MD_{+3}]) × (limb [dominant, non-dominant]) was used to identify the effect of the soccer match on hamstring MVIC torque and RTD, while a one-way repeated measures ANOVA (time [MD, MD_{+1}, MD_{+2}, MD_{+3}]) was used to identify soccer match-induced changes in serum biomarkers. Post hoc Bonferroni-corrected paired t tests were conducted to identify significant differences from baseline values. Percentage change with 95% CIs were also calculated. ES were calculated using Hedges' *g* (*g*), considering the following thresholds: (<0.20), trivial; (0.20 < 0.59), small; (0.60 < 1.19), moderate; (1.20 < 1.99), large; and (>2.0), very large (Hopkins et al. 2009). The area under the curve (AUC) of hamstring neuromuscular markers and serum biomarkers of structural/oxidative damage over a defined time interval (i.e., from baseline to MD_{+3}) was calculated by the trapezoidal method. Associations between AUCs were analyzed using Pearson's or Spearman correlation coefficient (according to normality) considering the following thresholds: (<0.30), small; (0.30 < 0.51), moderate; (0.51 < 0.70), large; (0.71 < 0.90), very large; and (>0.90), extremely large (Hopkins et al. 2009). Data are presented as mean ± standard deviation (SD) and the level of significance was set at $p < 0.05$. Statistical analysis was performed with SPSS v.27.0.1.0 (SPSS Statistics, IBM Corp., Armonk, NY, USA).

Results

Soccer match loading

The players reached the following load parameters during the soccer match: TD = 10,463 ± 623 m; HSRd = 466 ± 257 m; Acc = 45 ± 21 accelerations over 3 m s^{-2}; Dec = 59 ± 24 decelerations over 3 m s^{-2}.

Hamstring's capacity to generate maximal and rapid force

Maximal voluntary isometric contraction (MVIC) torque

Hamstring MVIC torque measured by the *90°$_{hip}$:20°$_{knee}$ IPC test* showed an excellent test–retest reliability (ICC = 0.987, 95% CI = 0.977–0.993; %TE = 0.61, 95% CI = 0.41–0.81; SWC = 12.8 N m). As shown in Fig. 2, there was MVIC torque loss in both dominant and non-dominant hamstrings until MD_{+2} [$F(3,54) = 8.891$, $p < 0.001$, $n_p^2 = 0.331$] and recovered on MD_{+3}. There were no between-limb differences in the loss of hamstring MVIC torque during the post-match recovery period [$F(3,54) = 0.350$, $p = 0.789$].

Rate of torque development (RTD)

Reliability measures for $RTD_{0–50}$ (ICC = 0.750, 95% CI = 0.567–0.864; %TE = 10.72, 95% CI = 9.08–12.37; SWC = 30.7 N m s^{-1}), $RTD_{0–100}$ (ICC = 0.776, 95% CI = 0.611–0.878; %TE = 8.70, 95% CI = 6.94–10.45; SWC = 40.0 N m s^{-1}), $RTD_{0–250}$ (ICC = 0.832, 95% CI = 0.670–0.914; %TE = 5.56, 95% CI = 4.28–6.85; SWC = 33.2 N m s^{-1}) and RTD_{peak} (ICC = 0.712, 95% CI = 0.504–0.842; %TE = 9.53, 95% CI = 7.39–11.67; SWC = 75.8 N m s^{-1}) were moderate to good.

$RTD_{0–50}$ [$F(3,54) = 33.224$, $p < 0.001$, $n_p^2 = 0.649$] and $RTD_{0–100}$ [$F(3,54) = 17.754$, $p < 0.001$, $n_p^2 = 0.497$] were decreased in both dominant and non-dominant hamstring muscles until MD_{+3}, while $RTD_{0–250}$ was decreased until MD_{+2} [$F(3,54) = 7.894$, $p < 0.001$, $n_p^2 = 0.305$]. RTD_{peak} remained unchanged during the post-match recovery period. There were no between-limb differences in $RTD_{0–50}$, $RTD_{0–100}$, $RTD_{0–250}$ and RTD_{peak} during the post-match recovery period (Fig. 3).

Normalized RTD metrics by MVIC torque showed a decrease until MD_{+3} for the 0–50 ms [$F(3,54) = 15.767$, $p < 0.001$, $n_p^2 = 0.467$] and 0–100 ms [$F(3,54) = 6.492$, $p = 0.003$, $n_p^2 = 0.265$] intervals, while there were no significant changes for the 0–250 ms interval [$F(3,54) = 1.169$, $p = 0.330$] or RTD_{peak} [$F(3,54) = 1.882$, $p = 0.144$]. There were no between-limbs differences in any of the normalized RTD metrics by MVIC torque (Table 1).

Biochemical markers panel

Serum biomarkers

CK activity [$F(3,48) = 22.673$, $p < 0.001$, $n_p^2 = 0.586$], CK/ALT ratio [$F(3,51) = 35.786$, $p < 0.001$, $n_p^2 = 0.678$], CK/

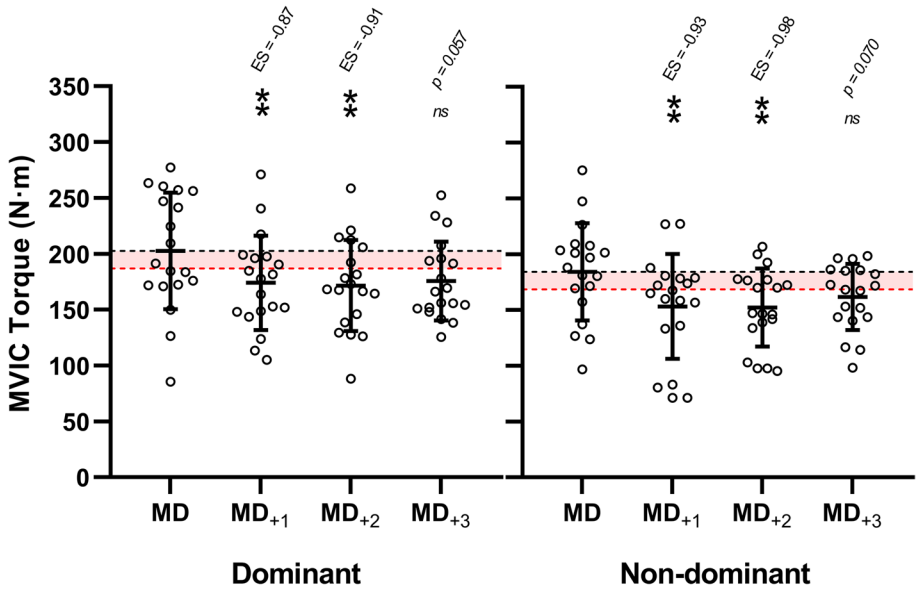

Fig. 2 Individual, mean ± SD values of maximal voluntary isometric contraction (MVIC) torque during post-soccer match recovery period. *, ** and *** indicate a significant difference from the baseline value at $p < 0.05$, $p < 0.01$ and $p < 0.001$, respectively. The range between dashed lines represents the smallest worthwhile change in MVIC torque measured through the *90°$_{hip}$:20°$_{knee}$ IPC test*. ES, effect size; MD, match day; ns, non-significant

AST ratio [$F(3,51) = 47.686$, $p < 0.001$, $n_p^2 = 0.737$] and sMtCK [$F(3,54) = 39.814$, $p < 0.001$, $n_p^2 = 0.689$] were increased until MD_{+3}. Similarly, MDA [$F(3,54) = 15.139$, $p < 0.001$, $n_p^2 = 0.457$] and circulating irisin [$F(3,54) = 16.300$, $p < 0.001$, $n_p^2 = 0.475$] were increased after the soccer match until MD_{+3} (Table 2). Correlations between serum enzyme levels and long-lasting reductions in neuromuscular markers are shown in Table S2.

Dietary intake

The nutritional intake of the participants was homogeneous and within normal ranges for each macro and micronutrient (total energy = 2261 ± 787 kcal day^{-1}; total carbohydrates = 219 ± 83 g day^{-1}; total protein intake = 1.65 ± 0.69 g kg^{-1} day^{-1}; total lipids = 101 ± 44 g day^{-1}). Detailed information on nutritional intake is summarized in Table S3.

Discussion

The recovery kinetics of hamstring's capacity to generate maximal and rapid force were monitored during 72 h after a soccer match. The findings of the research highlight that (1) early $RTD_{0–50}$ and $RTD_{0–100}$ monitoring, through the *90°$_{hip}$:20°$_{knee}$ IPC test*, is a cost-effective and non-exhaustive method for assessing soccer match-induced hamstring muscles residual fatigue and recovery; (2) loss of rapid force-generating capacity (i.e., early RTD) is accompanied by biomarkers of muscle damage and recovery; and (3) 72 h is not adequate time to recover soccer match-induced hamstring muscles residual fatigue.

Available scientific evidence is limited and contradictory on soccer match-derived hamstring muscles residual fatigue. So far, the assessment of hamstring's force-generating capacity using supine IPC test has shown both, moderate decreases to hamstring's force-generating capacity at MD_{+3} (Nedelec et al. 2014) and recovered hamstrings prior to MD_{+2} (Constantine et al. 2019; Springham et al. 2024), in professional soccer players. Similarly, we observed a significant decrease in both dominant and non-dominant hamstring MVIC torque until MD_{+2} (ES = −0.91 and ES = −0.98, respectively), which was recovered by MD_{+3}, tested by the *90°$_{hip}$:20°$_{knee}$ IPC test*. Reduced performance in the *90°$_{hip}$:20°$_{knee}$ IPC test* accounts for a loss of hamstring's force-generating capacity. Specifically, the positioning of the hamstring muscles during the test is comparable to the positioning during the late swing phase of the running gait cycle (i.e., extensive hip flexion with reduced knee flexion) (Thelen et al. 2005), reaching the peak electromyographic activity (Higashihara et al. 2010; Kellis et al. 2017) and length–tension relationship in the BF_{LH} (Chleboun et al. 2001), together with a greater moment arm (Thelen et al. 2005), compared to the semitendinosus or semimembranosus muscles. Therefore, loss of performance in the *90°$_{hip}$:20°$_{knee}$ IPC test* is suggestive of hamstring's (and most likely the biceps femoris) residual fatigue.

Besides, rapid force production provides a reliable monitoring of fatigue over time (Ripley et al. 2023). However, to our knowledge, the monitoring of hamstring's rapid force production during the 72-h recovery period after a soccer match is described for the first time. Interestingly, both dominant and non-dominant hamstring muscles showed significant decreases in early $RTD_{0–50}$ (ES = −1.25 and ES = −1.21, respectively) and $RTD_{0–100}$ (ES = −1.08 and ES = −1.01, respectively) until MD_{+3}, whereas RTD_{peak} remained unchanged. Similarly, normalized RTD metrics by MVIC torque showed significant and moderate decreases

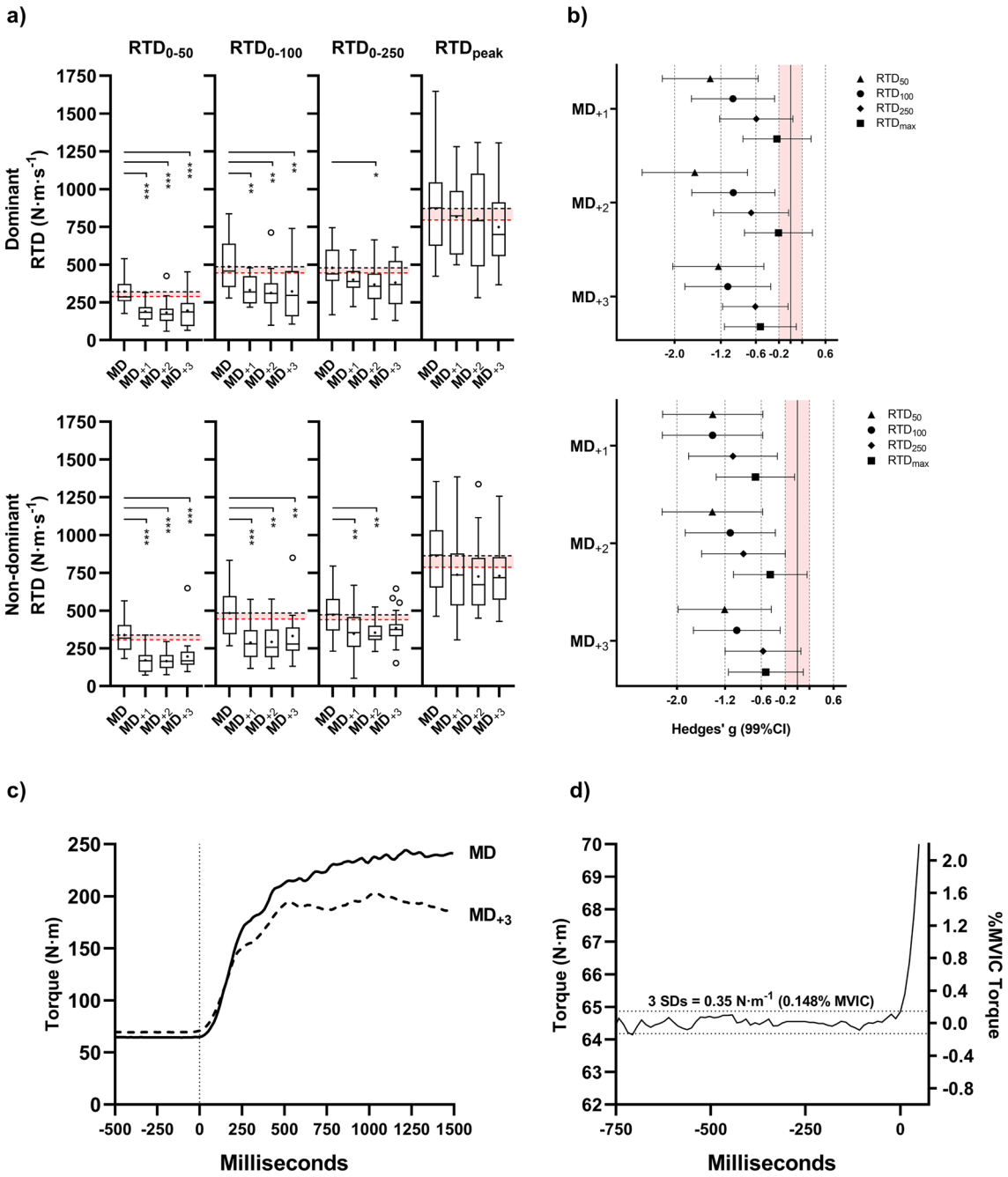

Fig. 3 Tukey boxplot showing the rate of torque development during post-soccer match recovery period in dominant and non-dominant hamstring muscles (**a**). The upper and lower edge of the box contains the interquartile range of the data set, while "whiskers" indicate the minimum and maximum data values (outliers are illustrated as open dots). The range between dashed lines represents the smallest worthwhile change for each RTD metric. **b** Reports rate of torque development standardized differences (Hedges' *g*) comparing baseline values with post-24 h (MD_{+1}), post-48 h (MD_{+2}) and post-72 h (MD_{+3}) in dominant and non-dominant hamstring muscles. Error bars indicate uncertainty in the true mean changes with a 99% confidence interval. **c** Illustrates the torque-time curves of volunteer 17, on MD and MD_{+3}, while **d** illustrates baseline-torque recordings prior to contraction onset. Dashed lines represent ± 3 standard deviations (SD) of the mean baseline noise amplitude

until MD_{+3} for early time intervals (i.e., 0–50 ms and 0–100 ms), demonstrating that hamstring muscles lost the ability to express rapid force-generating capacity (Maffiuletti et al. 2016). Therefore, similar to previous research (Jenkins et al. 2014; Peñailillo et al. 2015), there were longer-lasting decreases in early RTD metrics, compared to MVIC torque assessment. In addition, there were greater decreases in early RTD than late RTD, supporting the findings of a

Table 1 Normalized rate of torque development (RTD) metrics by maximal voluntary isometric contraction (MVIC) torque during post-soccer match recovery period

Metric	MD	MD_{+1}	MD_{+2}	MD_{+3}
$RTD_{0–50}$ (%MVIC s^{-1})				
D				
Mean ± SD	164.7 ± 54.4	113.7 ± 40.3***	111.8 ± 50.6***	112.0 ± 52.2***
%Δ (95% CI)		−31.0 (−42.7 to −19.2)	−32.1 (−46.9 to −17.3)	−32.0 (−47.2 to −16.8)
ES (95% CI)		−1.03 (−1.56 to −0.48)	−1.35 (−1.96 to −0.73)	−0.92 (−1.43 to −0.39)
ND				
Mean ± SD	193.0 ± 72.5	116.7 ± 50.0***	117.5 ± 56.8***	120.7 ± 61.6***
%Δ (95% CI)		−39.5 (−52.0 to −27.0)	−39.1 (−53.3 to −25.0)	−37.3 (−52.8 to −22.1)
ES (95% CI)		−1.09 (−1.64 to −0.53)	−0.93 (−1.45 to −0.40)	−0.88 (−1.39 to −0.36)
$RTD_{0–100}$ (%MVIC s^{-1})				
D				
Mean ± SD	244.4 ± 61.9	195.8 ± 51.1**	189.0 ± 83.0*	183.7 ± 85.7**
%Δ (95% CI)		−19.9 (−30.0 to −9.9)	−22.7 (−39.0 to −6.3)	−24.8 (−41.7 to −8.0)
ES (95% CI)		−0.65 (−1.12 to −0.16)	−0.77 (−1.26 to −0.26)	−0.77 (−1.26 to −0.26)
ND				
Mean ± SD	268.4 ± 72.8	194.7 ± 79.9**	206.1 ± 100.3*	203.9 ± 79.5**
%Δ (95% CI)		−27.5 (−41.8 to −13.1)	−23.2 (−42.0 to −5.2)	−24.0 (−38.3 to −9.8)
ES (95% CI)		−0.71 (−1.19 to −0.22)	−0.50 (−0.96 to −0.26)	−0.75 (−1.24 to −0.25)
$RTD_{0–250}$ (%MVIC s^{-1})				
D				
Mean ± SD	238.6 ± 48.2	233.8 ± 40.2	219.1 ± 76.1	218.0 ± 70.5
%Δ (95% CI)		−2.0 (−10.1 to 6.1)	−8.2 (−23.5 to 7.2)	−8.6 (−22.9 to 5.6)
ES (95% CI)		−0.09 (−0.52 to 0.34)	−0.26 (−0.70 to 0.18)	−0.29 (−0.72 to 0.26)
ND				
Mean ± SD	258.0 ± 47.0	225.7 ± 64.2	242.2 ± 61.6	236.2 ± 55.3
%Δ (95% CI)		−12.5 (−24.5 to −0.5)	−6.1 (−17.6 to 5.4)	−8.5 (−18.8 to 1.9)
ES (95% CI)		−0.37 (−0.81 to 0.08)	−0.21 (−0.64 to 0.23)	−0.31 (−0.75 to 0.14)
RTD_{peak} (%MVIC s^{-1})				
D				
Mean ± SD	435.4 ± 112.0	476.1 ± 128.6	481.3 ± 173.5	427.6 ± 128.4
%Δ (95% CI)		9.4 (−4.9 to 23.6)	10.5 (−8.7 to 29.8)	−1.8 (−16.0 to 12.5)
ES (95% CI)		0.42 (−0.04 to 0.86)	0.30 (−0.15 to 0.74)	−0.07 (−0.50 to 0.36)
ND				
Mean ± SD	473.0 ± 80.2	494.2 ± 114.8	498.3 ± 178.6	454.4 ± 118.5
%Δ (95% CI)		4.5 (−7.2 to 16.2)	5.3 (−12.9 to 23.5)	−3.9 (−16.0 to 8.1)
ES (95% CI)		0.16 (−0.27 to 0.59)	0.14 (−0.29 to 0.57)	−0.14 (−0.57 to 0.29)

Data are expressed as mean ± SD

95% CI 95% confidence interval, *%Δ* delta percentage change, *D* dominant, *ES* effect size, *ND* non-dominant, *SD* standard deviation

*, **, and *** indicate a significant difference from the baseline value at $p < 0.05$, $p < 0.01$ and $p < 0.001$, respectively

recent comprehensive review (D'Emanuele et al. 2021). The determinants of the force rise time during an isometric contraction are the magnitude and shortening velocity of the muscle, which depends on two simultaneous muscle physiological processes and properties (Edman and Josephson 2007). First, 60% of the time to reach 50% MVIC (i.e., early RTD) is mediated by activation processes, including both calcium ion (Ca^{2+}) release, diffusion and binding, as well as changes in filament organization to allow cross-bridge binding, while the remaining 40% of the time is influenced by the compliance of muscle elastic components (Edman and Josephson 2007). Eccentric contraction-induced muscle strains during repeated late swing phases of the running gait cycle lead to the disruption of elastic components (e.g., muscle membranes) causing sarcolemma permeability that affects Ca^{2+} homeostasis and, consequently, delays in

Table 2 Soccer match-induced biochemical alterations during the 72-h recovery period

Marker	MD	MD_{+1}	MD_{+2}	MD_{+3}
CK (IU L^{-1})				
Mean ± SD	203.6 ± 100.3	1220.2 ± 799.2***	741.5 ± 488.1**	456.8 ± 218.6**
%Δ (95% CI)		499.3 (297.5–701.2)	264.2 (141.0–387.5)	124.4 (69.2–179.6)
ES (95% CI)		1.26 (0.62–1.88)	1.13 (0.51–1.72)	1.11 (0.50–1.70)
CK/ALT (AU)				
Mean ± SD	26.2 ± 25.8	59.2 ± 33.5***	43.7 ± 21.1***	33.5 ± 15.9**
%Δ (95% CI)		126.3 (62.6–190.0)	66.8 (26.8–106.8)	27.9 (−2.3 to 58.2)
ES (95% CI)		1.58 (0.88–2.26)	1.38 (0.73–2.02)	0.94 (0.38–1.48)
CK/AST (AU)				
Mean ± SD	12.4 ± 10.5	48.4 ± 34.0***	43.7 ± 21.1***	33.5 ± 15.9**
%Δ (95% CI)		289.3 (153.4–425.2)	113.8 (53.1–174.5)	65.6 (20.9–110.4)
ES (95% CI)		1.78 (1.02–2.51)	1.42 (0.75–2.06)	1.03 (0.45–1.58)
sMtCK (ng ml^{-1})				
Mean ± SD	18.5 ± 15.2	100.4 ± 48.9***	71.9 ± 32.4***	46.6 ± 26.5**
%Δ (95% CI)		443.6 (316.0–571.2)	289.3 (204.8–373.9)	151.9 (82.7–221.1)
ES (95% CI)		1.65 (0.95–2.34)	1.59 (0.90–2.25)	0.92 (0.38–1.44)
MDA (ng ml^{-1})				
Mean ± SD	836.8 ± 105.7	1197.7 ± 360.4**	1358.5 ± 364.5***	1384.8 ± 479.5**
%Δ (95% CI)		43.1 (22.4–63.9)	62.3 (41.3–83.3)	65.5 (37.9–93.1)
ES (95% CI)		1.04 (0.48–1.59)	1.37 (0.74–1.99)	1.13 (0.55–1.69)
Irisin (ng ml^{-1})				
Mean ± SD	7.198 ± 1.223	9.410 ± 2.560**	10.186 ± 2.532***	10.505 ± 2.062***
%Δ (95% CI)		30.7 (13.6–47.9)	41.5 (24.6–58.5)	45.9 (32.1–59.7)
ES (95% CI)		0.86 (0.33–1.37)	1.36 (0.73–1.97)	1.61 (0.92–2.29)

Data are expressed as mean ± SD

95% CI 95% confidence interval, *%Δ* delta percentage change, *AU* arbitrary units, *CK* creatine kinase, *CK/ALT* CK to alanine aminotransferase (ALT) ratio, *CK/AST* CK to aspartate aminotransferase (AST) ratio, *ES* effect size, *IU* international units, *MD* match day, *MDA* malondialdehyde, *SD* standard deviation, *sMtCK* sarcomeric mitochondrial creatine kinase

*, **, and *** indicate a significant difference from the baseline value at $p < 0.05$, $p < 0.01$ and $p < 0.001$, respectively

muscle activation processes (e.g., excitation–contraction coupling process failures) (Hyldahl and Hubal 2014). Thus, soccer match-induced long-lasting early RTD reductions to the hamstring muscles likely result from (1) excitation–contraction uncoupling as a result of loss of Ca^{2+} homeostasis and (2) reduced stiffness of cellular force transmission associated with impaired myocyte integrity.

Accordingly, elastic components and structural damage in the volunteers were evidenced by an increase in CK activity until MD_{+3} (ES = 1.11). In addition, elevated CK/ALT (ES = 0.94) and CK/AST ratios (ES = 1.03) until MD_{+3}, together with CK activity, is an indicative of a disruption in cellular and sarcomeres integrity, as well as permeability of the sarcolemma (Hyldahl and Hubal 2014; Lim 2020). Sarcolemma impairment-derived Ca^{2+} accumulation disrupts the mitochondrial matrix, finally resulting in rupture of the mitochondrial outer membrane and intermembrane proteins efflux (i.e., sMtCK) (Görlach et al. 2015). In this sense, sMtCK, a muscle mitochondria damage marker that is increased after sprint-based exercises (Cosio et al. 2024), was also increased in the volunteers until MD_{+3} (ES = 0.92). Impaired mitochondria-derived free radicals release to the myocyte and bloodstream increases oxidative by-products (e.g., MDA). In this regard, there were moderate increases in MDA until MD_{+3} (ES = 1.13). Coupled with increases in CK activity, MDA further supports sarcolemma impairment and protein leakage into the bloodstream (Fatouros et al. 2010). The aforementioned mechanical and oxidative damage to myocyte and mitochondria integrity leads to an upregulation of the signaling pathway for irisin cleavage (Liu et al. 2022). The myokine irisin provides protection against inflammation, oxidative and apoptosis processes (Askari et al. 2018), but also plays a key role in musculoskeletal disorders by upregulating the expression of pro-myogenic factors, stimulating satellite cell activation and ameliorating protein degradation (Reza et al. 2017). Accordingly, the findings of the present study revealed a large increase ranging from 32.1 to 59.7% in circulating irisin on MD_{+3} (ES = 1.61), supporting

the previous results after sprint-based exercises (Cosio et al. 2024). Indeed, the findings showed significant moderate correlations between irisin and CK/AST ratio ($r = 0.73$), damage to muscle mitochondria ($r = 0.53$), MVIC torque loss ($r = 0.49$), $RTD_{0–50}$ loss ($r = 0.52$) and $RTD_{0–100}$ loss ($r = 0.52$). Thereby, exercise-induced muscle damage might trigger irisin cleavage to regulate signaling processes during muscle recovery. Taken together, the upregulation of serum biomarkers support the loss of cellular integrity and, consequently, a reduced stiffness of cellular force transmission which might explain the long-lasting early RTD decrements of the hamstrings (Peñailillo et al. 2015).

While there is an International Olympic Committee consensus to intersperse soccer matches by at least 96 h (Soligard et al. 2016), the governing sports authorities and professional leagues continue to schedule games every 72 h (Brito 2017). However, a recent comprehensive study examined the time course of HSI risk factors through a soccer-contextualized multifactorial approach (e.g., histology, range of motion, lumbo-pelvic control), revealing that HSI risk factors are not recovered within 72 h after a soccer match (Carmona et al. 2024a). Overall, combining the existing evidence with the findings hereby presented on decreased hamstring muscle capacity to generate rapid force (i.e., early RTD) until MD_{+3}, together with sustained biochemical responses, we postulate that a soccer match impairs hamstring muscles for at least 72 h. Accordingly, from a perspective for professional praxis, the findings of this study support the need for practitioners to improve the recovery of hamstrings rapid force-generating capacity, using individualized recovery strategies based on regular screening of hamstrings early RTD metrics. Thus, by assessing early RTD metrics, practitioners would be able to individualize recovery strategies after a soccer match according to the hamstrings condition of each player, thereby reducing HSI risk and improving performance, especially during periods of congested schedules. Also, it would be worthwhile to increase hamstrings rapid force production to prevent subsequent decline after a soccer match, through hamstring strengthening training programs (Buckthorpe et al. 2019). Therefore, we recommend the use of the $90°_{hip}$:$20°_{knee}$ *IPC test*, which is a non-exhaustive, fast and cost-effective test to monitor the recovery kinetics of force-generating capacity and, especially, early RTD metrics of the hamstring muscles after multiple sprints-based sports, to balance optimizing training and muscle recovery to decrease HSI risk.

Limitations

The study also presented some limitations to highlight when interpreting the results. First, although the used strain gauge provides a low baseline noise amplitude and is reliable for rapid force recording and analysis (Courel-Ibáñez et al. 2020), the load cell used did not provide a 1000 Hz sampling frequency, as recommended by (Maffiuletti et al. 2016). However, it has been recently demonstrated that lowering the sampling frequency (i.e., 1000 to 100 Hz) reduces the coefficient of variation and increases the absolute reliability during unilateral IPC tests (e.g., $90°_{hip}$:$20°_{knee}$ *IPC test*) (Ripley et al. 2024c). In addition, given that early RTD metrics together with biochemical assessments were affected during the 72-h recovery period after the soccer match, further research conducting additional biochemical and neuromuscular measurements (e.g., +96 h, until complete recovery) are warranted to comprehensively understand soccer match-induced hamstrings residual fatigue and recovery.

Conclusion

The assessment of early $RTD_{0–50}$ and $RTD_{0–100}$ through the $90°_{hip}$:$20°_{knee}$ *IPC test* is a non-exhaustive screening tool to monitor muscle recovery during soccer match-induced hamstrings residual fatigue. Overall, hamstrings muscle recovery is not fully completed within 3 days after a soccer match, owing to neuromuscular alterations associated with biomarkers of muscle damage and recovery.

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Data availability The datasets generated/analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare that the research was conducted in the absence of any commercial, financial, or non-financial relationships that could be construed as a potential conflict of interest.

Consent to participate Informed consent was obtained from all individual participants included in the study.

Ethics approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. The study was approved by the Ethics Committee for Clinical Research of the Catalan Sports Council (Generalitat de Catalunya) (No. 037/CEICGC/2021).

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